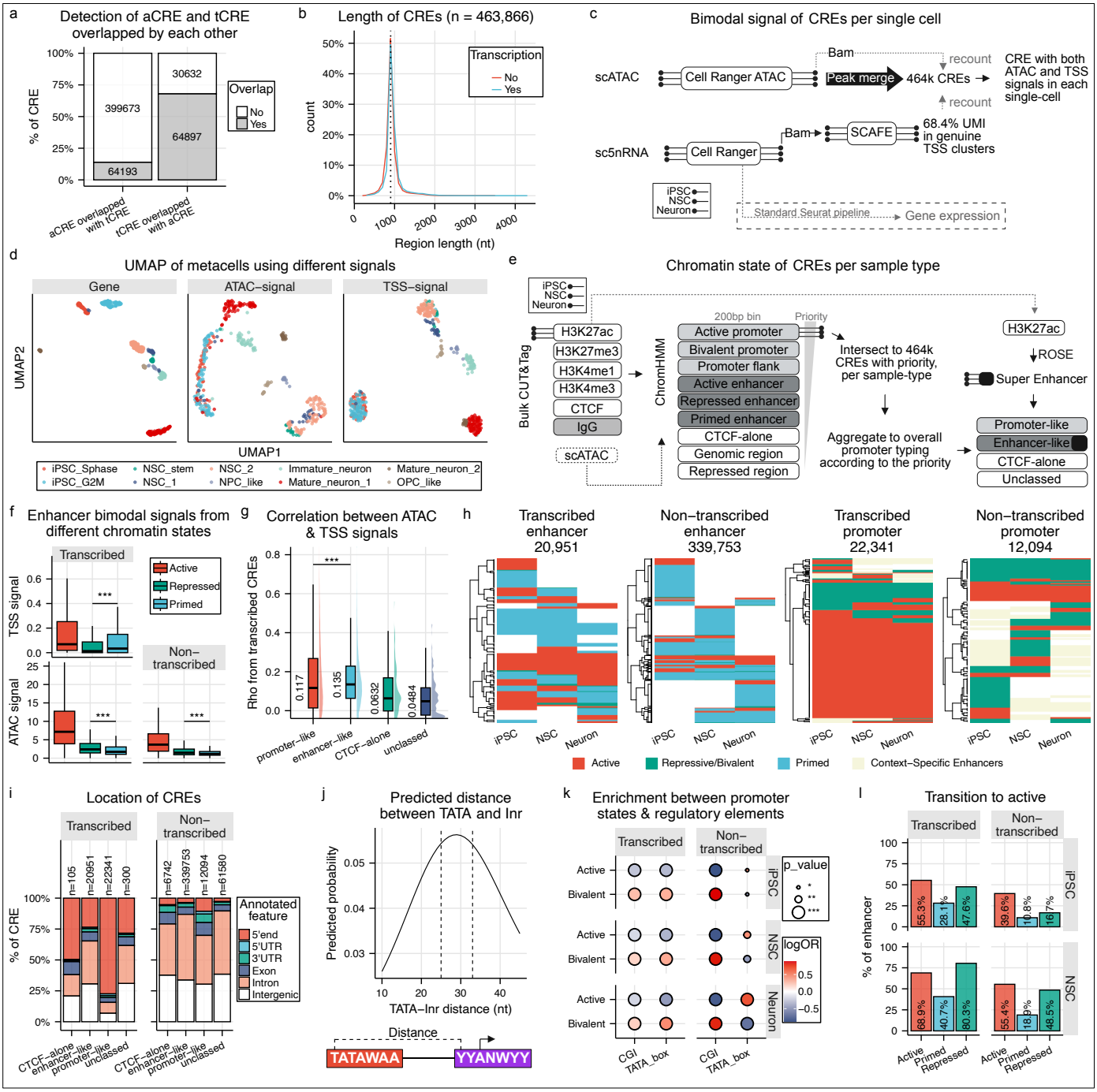


Extended Figure 2 | Bimodal signals and genomic features of the defined CREs.

a, Identification of aCRE derived from merged ATAC peaks and tCRE derived from SCAFE, and the percentage of them being overlapped with each other. **b**, Distribution of sequence length for all CREs, grouped as transcribed and non-transcribed. **c**, Schematic illustrating the quantification of bimodal signals at CREs using the scATAC-seq (accessibility) and sc5nRNA-seq (transcription start sites (TSS)) data. **d**, UMAP visualizations of metacells based on gene-signal, ATAC-signal and TSS-signal. **e**, Schematic showing the annotation of chromatin states of CREs based on bulk CUT&Tag from iPSC, NSC and Neuron using ChromHMM and the identification of super enhancer using ROSE. **f**, CRE Bimodal signals from different chromatin states, grouped in transcribed or non-transcribed and promoter or enhancer. Signals were aggregated from three samples. **g**, Spearman correlation between CRE ATAC and TSS signals across metacells in different classes of promoter-type. **h**, Heatmaps displaying the chromatin states transitions across the three samples. The CREs were stratified into transcribed enhancers, non-transcribed enhancers, transcribed promoters and non-transcribed promoters. **i**, Genomic distribution of CREs relative to annotated protein-coding gene coordinates. **j**, Predicted distance between TATA-box motif and Initiator motif from TATA-box mediated transcription sites (TSS tag >=3 inside the Initiator motif) with 25-33 nucleotides reaching 95% of maximum predicted probability. Schematics at the bottom show the positions where the distance measured. **k**, Enrichment between regulatory elements (CGI and TATA box) and chromatin states of promoter-like CREs. A subset of promoter-like CREs that are linked to annotated genes was used. **l**, Percentage of enhancer-like CREs with active chromatin state in at least one of the three samples across differentiation. Enhancers were grouped according to their chromatin states from iPSC and NSC separately. Fisher's exact test was used to calculate odds ratios (OR). $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***), n.s. = not significant. For (**f**) and (**g**), statistical comparisons were performed using a two-sided Wilcoxon test.